Integer programs and branch and bound



Optimization · Level 4 · Lesson 2

In this lesson you will

- Explain why rounding an LP can fail, and why the relaxation is a bound.
- Follow branch and bound: branch, bound, prune, and the gap.
- Model whole units and a fixed cost in SCIP.

Key ideas

- Relaxation: a bound, not an answer
- Branch on a fraction; prune what can’t win
- Gap = (best - bound) ÷ best
- Yes/no variables model logic and fixed costs

Activity: The bus tree by hand

1. Root (5 min): Solve the relaxation: 2.13 buses.
2. Branch (8 min): At most 2 buses, or at least 3.
3. Prune (7 min): Find 1,400; cut the 3-bus branch.
4. Fee (5 min): Add the booking fee and redo.

Work in pairs: one draws the tree, the other checks each relaxation. Then run the same model in the lesson's code step and compare.

Let’s discuss

1. Why can rounding an LP answer break a constraint?
2. Why does a smaller big-M make branch and bound faster?
3. When is a 3% gap good enough, and who decides?

Check yourself

- A relaxation costs 900 and your best plan 1,000. Write the gap, and what it promises.